APPLIED MACHINE LEARNING TECHNIQUES · MODULE 4

# Unsupervised Learning, Made Simple

Clustering · Association Rules · Evaluation Metrics — explained with everyday analogies, worked math, and zero jargon.

K-Means · GMM

Market Baskets

Cluster Quality

ROADMAP

# What We'll Cover Today

- What unsupervised learning is (and isn't)
- How it differs from supervised learning
- Real-world uses: marketing, healthcare, security
- Anomaly detection: point, contextual, collective
- Customer segmentation in business
- K-Means clustering + the Elbow Method
- Hierarchical clustering & dendrograms

- Gaussian Mixture Models & soft clustering
- Association Rule Learning
- Market basket analysis & rule mining
- Support, Confidence, and Lift — by hand
- Apriori, FP-Growth, and Eclat algorithms
- Cosine similarity, step by step
- Judging clusters: Silhouette & Davies-Bouldin

GOALS

# By the End, You Will Be Able To…

- Tell apart supervised and unsupervised learning — instantly, with examples.
- Spot where unsupervised learning quietly runs the real world around you.
- Cluster data using K-Means, Hierarchical methods, and Gaussian Mixture Models.
- Mine “bought-together” rules with Apriori, FP-Growth, and Eclat.
- Score cluster quality using Silhouette Score and the Davies-Bouldin Index.

PART 1

# Introduction to Unsupervised Learning

01

- → The core idea — learning without an answer key
- → Everyday analogies: jigsaw puzzles, fruit baskets, and bikes
- → The process flow, and the three big approaches
- → Supervised vs. unsupervised — side by side

THE CORE IDEA

# What Is Unsupervised Learning?

**Unsupervised learning** is machine learning on data that carries **no labels and no right answers**. The model isn't told what to find — it discovers hidden patterns, groups, and relationships on its own.

### Analogy: a jigsaw with no box photo

You've lost the box, so nobody can tell you what the final picture looks like. You still make progress: sky-blue pieces in one pile, straight edges in another, green textures in a third. You grouped by similarity — exactly what unsupervised learning does.

### Contrast: studying with an answer key

Supervised learning is like practicing with solved papers — every question comes with the correct answer, and you learn the mapping. Unsupervised learning is handed only the questions and asked: “what structure do you see?”

INTUITION CHECK

# Can You Guess the Fruit From Features Alone?

No labels are given — only color, shape, and size. Your brain clusters them anyway. That instinct is the algorithm.

| Color | Shape | Size | Your brain says... |
|---|---|---|---|
| Yellow | Long, curved cylinder | Big | **Banana** |
| Red | Round, dip on top | Big | **Apple** |
| Green / purple | Small ovals in a bunch | Small | **Grapes** |

**Key point:** you never saw a label saying “banana”. You formed groups purely from shared features — and a name for each group came *after* the grouping. Unsupervised models work in exactly this order: cluster first, interpret later.

HOW IT WORKS

# The Unsupervised Learning Process Flow

### 1 · Unlabeled data

A pile of raw examples — no tags, no categories, no answers attached.

### 2 · Train the model

The algorithm measures similarity between examples (distance, density, probability).

### 3 · Groups emerge

Similar items pull together into clusters; dissimilar items drift apart.

### 4 · New data arrives

A fresh example is assigned to whichever existing group it resembles most.

**Note:** “training data” here just means a collection of information **without any labels** — the model is never shown a correct answer at any point.

WORKED EXAMPLE

# Sorting Vehicles Without Being Told What They Are

1. Feed in a mixed set of two-wheelers — bicycles, scooters, sports bikes — described only by features: size, top speed, engine, color.

2. The system notices natural groupings: small + pedal-powered, medium + step-through, large + high speed — and forms categories on its own.

3. Show it a brand-new vehicle: it compares features and drops it into the most similar existing group.

**The groups depend entirely on the attributes you choose.** Feed in color → you get red vs. green piles. Feed in speed → racing vs. commuter piles. Same data, different lens, different clusters. Feature choice is the steering wheel of unsupervised learning.

THE TOOLBOX

# Three Families of Unsupervised Learning

### Clustering

Group similar things together.

Like sorting a laundry basket: whites, colors, delicates — no one labeled the clothes, the similarity did the work.

Examples: K-Means, GMM, Hierarchical.

### Association Rules

Find what goes together.

Like noticing chai is almost always ordered with biscuits — a “if A, then probably B” pattern.

Examples: Apriori, FP-Growth, Eclat.

### Dimensionality Reduction

Compress many features into few.

Like a movie trailer: 2 minutes that keep the story of 2 hours. Less data, same essence.

Examples: PCA, t-SNE, Autoencoders.

*This module focuses on the first two — clustering and association rules.*

SIDE BY SIDE

# Supervised vs. Unsupervised Learning

| Aspect | Supervised | Unsupervised |
| --- | --- | --- |
| **Data** | Labeled — every example has an answer | Unlabeled — raw examples only |
| **Goal** | Predict outcomes (classify / regress) | Discover structure (clusters / rules) |
| **Human effort** | Heavy — someone must label the data | Light — no labeling needed |
| **Algorithms** | Linear Regression, SVM, Decision Trees | K-Means, PCA, Hierarchical, Apriori |
| **Typical use** | Spam detection, fraud prediction | Customer segments, topic discovery |

**One-line memory hook:** supervised = exam with an answer key; unsupervised = detective work with no key, only clues.

# Same Data, Different Question

### Supervised

You get pairs: $(x_1, y_1), (x_2, y_2), \ldots$

Learn a function $f : X \rightarrow y$ that maps inputs to known answers.

### Unsupervised

You get only inputs: $x_1, x_2, \ldots, x_n$

Learn the **structure of X itself** — groups, co-occurrence rules, or a compressed form.

**Concrete example — same spreadsheet of customers:**

**Supervised question:** "Given past churners (labels), will THIS customer churn?"

**Unsupervised question:** "With no labels at all, what natural customer groups exist in here?"

PART 2

02

# Where It's Used in the Real World

- Five everyday applications you already interact with
- Deep dives: healthcare, marketing, cybersecurity
- The honest list of challenges

APPLICATIONS

# It's Already All Around You

### News grouping

Google News bundles hundreds of articles about the same event — nobody labels them; similarity of words does it.

### Computer vision

Grouping visually similar images or regions — e.g., photo apps clustering faces without knowing names.

### Medical imaging

Separating tissue types in MRI/CT scans by pixel patterns, helping flag unusual regions.

### Anomaly detection

Learning what “normal” looks like, then raising a flag when something deviates — fraud, intrusions, faults.

### Recommendations

“Customers like you also bought…” — grouping users by behavior, not by declared identity.

USE CASE 1

# Healthcare — Finding Patient Groups No One Defined

Hospitals cluster patients by patterns in vitals, lab results, and history — **without predefined risk labels** — and discover risk segments the staff never explicitly created.

| Cluster found | Pattern (BP, cholesterol) | What doctors do with it |
|---|---|---|
| **Group A** | Low BP, low cholesterol | Routine annual checkups |
| **Group B** | Borderline values, drifting upward | Lifestyle program + 6-month review |
| **Group C** | High BP and high cholesterol together | Priority screening & medication review |

**The win:** the “high-risk” label was never in the data — the cluster surfaced it, and humans named it afterwards.

USE CASE 2

# Marketing — Customer Segmentation

Group customers by behavior — purchase frequency, average spend, browsing — **with no predefined segments**. The algorithm draws the boundaries; marketing names them.

| Customer | Visits / month | Avg spend (₹) | Cluster → nickname |
|---|---|---|---|
| C1 | 13 | 3,100 | **Frequent big spenders** |
| C2 | 2 | 420 | **Occasional shoppers** |
| C3 | 7 | 1,100 | **Price-sensitive regulars** |
| C4 | 14 | 3,250 | **Frequent big spenders** |

**Payoff:** send VIP perks to big spenders, discount coupons to the price-sensitive — same budget, far better conversion.

USE CASE 3

# Cybersecurity — Catching Attacks Nobody Has Seen

The model learns what **normal** user and network behavior looks like. Anything that lands far from that normal cloud gets flagged — **no labeled attack examples required.**

### Normal looks like…

Sessions of 80–120 seconds moving 400–600 KB, clustered tightly together during office hours. Dense, boring, predictable — exactly what we want.

### Suspicious looks like…

A 3 a.m. session moving 950 KB in 30 seconds, sitting far outside the dense cloud. The model has never seen this attack before — distance alone betrays it.

**Why this matters:** supervised systems only catch attacks they were trained on; unsupervised systems can catch attacks invented yesterday.

HONEST LIMITATIONS

# Five Challenges of Unsupervised Learning

**Computational cost:** Comparing everything with everything is expensive — big data makes it heavier.

**Longer training time:** No answer key means more exploration before the structure settles.

**Risk of wrong results:** With no ground truth, the model can group things confidently — and wrongly.

**Humans must validate:** Someone has to inspect clusters and decide if they actually mean anything.

**Low transparency:** It's often unclear why two items landed in the same cluster — explainability suffers.